Q proves that everything below numeral n+1 is one of the numerals 0 through n

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-incompleteness-representability-in-q/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both cases are complete axiom-level derivations; no step is delegated

Editorial note. The label lem:less-nsucc names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Non-negativity lemma for Q. Same unit, \ollabel{lem:less-zero} (statement lines 1392-1395). Closes the base case: the Q_8 chain would otherwise force c < 0, which the stub forbids.

Source reference. Original lemma lem-less-zero: sources/olp-incompleteness-representability-in-q/pinned-original-source.tex, source lines 1392–1395. Prerequisite pointer invoked in the counted proof; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *For every natural number n,*

$$\mathrm{Th}Q \vdash \forall x\,(x < \overline{n+1} \to (x = \mathsf{0} \vee \cdots \vee x = \overline{n})).$$

Proof. We use induction on n. Let us consider the base case, when $n = 0$. In that case, we need to show $a < \overline{1} \to a = \mathsf{0}$, for arbitrary a. Suppose $a < \overline{1}$. Then by the defining axiom for $<$, we have $\exists y\,(y' + a) = \mathsf{0}'$ (since $\overline{1} \equiv \mathsf{0}'$).

Suppose b has that property, i.e., we have $(b' + a) = \mathsf{0}'$. We need to show $a = \mathsf{0}$. By axiom Q_3, we have either $a = \mathsf{0}$ or that there is a c such that $a = c'$. In the former case, there is nothing to show. So suppose $a = c'$. Then we have $(b' + c') = \mathsf{0}'$. By axiom Q_5 of ThQ, we have $(b' + c)' = \mathsf{0}'$. By axiom Q_1,

we have $(b' + c) = \mathsf{0}$. But this means, by axiom Q_8, that $c < \mathsf{0}$, contradicting lem-less-zero.

Now for the inductive step. We prove the case for $n + 1$, assuming the case for n. So suppose $a < \overline{n+2}$. Again using Q_3 we can distinguish two cases: $a = \mathsf{0}$ and for some b, $a = b'$. In the first case, $a = \mathsf{0} \vee \cdots \vee a = \overline{n+1}$ follows trivially. In the second case, we have $b' < \overline{n+2}$, i.e., $b' < \overline{n+1}'$. By axiom Q_8, for some c, $(c' + b') = \overline{n+1}'$. By axiom Q_5, $(c' + b)' = \overline{n+1}'$. By axiom Q_1, $(c' + b) = \overline{n+1}$, and so $b < \overline{n+1}$ by axiom Q_8. By inductive hypothesis, $b = \mathsf{0} \vee \cdots \vee b = \overline{n}$. From this, we get $b' = \mathsf{0}' \vee \cdots \vee b' = \overline{n}'$ by logic, and so $a = \overline{1} \vee \cdots \vee a = \overline{n+1}$ since $a = b'$. □